

Data Taming and Prediction (DTP) - Comprehensive Study Guide

This guide is designed to help you master the core concepts and practical skills required for the Data Taming and Prediction exam. It synthesizes key patterns from the practice exam and solutions, focusing on data wrangling, statistical interpretation, and predictive modeling using R.

1. Data Wrangling and Tidy Data Principles

The foundation of data analysis in R is the **Tidy Data** concept. Data is considered “tidy” when each variable is a column, each observation is a row, and each value is a cell.

Key R Functions

Function	Purpose
<code>pivot_longer()</code>	Converts “wide” data to “long” format by collapsing multiple measurement columns into two: a name column and a value column.
<code>group_by()</code>	Groups data by one or more variables for subsequent calculations.
<code>summarise()</code>	Reduces multiple rows to a single summary value (e.g., <code>sum()</code> , <code>mean()</code>).
<code>mutate()</code>	Creates or modifies columns (e.g., transforming a variable or stripping strings with <code>str_remove()</code>).

Exam Tip: Why Tidy Data?

You may be asked why long format is preferred. The answer is usually that it simplifies mapping variables to visual aesthetics (like `color` or `fill`) in `ggplot2` and allows for easier grouped analysis.

2. Statistical Interpretation and Visualization

The exam tests your ability to interpret statistical outputs and create appropriate visualizations.

Normality Checks

Before performing many statistical tests (like t-tests), you must check if the data follows a normal distribution.

- **Shapiro-Wilk Test:** If the p-value is **greater than 0.05**, we do not reject the null hypothesis of normality (the data is likely normal).
- **Q-Q Plots:** If the points fall closely along the diagonal reference line, the data is approximately normal. Departures at the “tails” (ends) suggest non-normality.

Hypothesis Testing (T-Tests)

- **Null Hypothesis (H_0):** There is no difference between the means of the two groups.
- **Alternative Hypothesis (H_A):** There is a statistically significant difference between the means.
- **Decision Rule:** If $p < 0.05$, reject H_0 .

Visualization with `ggplot2`

For comparing categories (e.g., rental types across locations), use a **grouped bar plot**:

```
ggplot(data, aes(x = Category, y = Value, fill = Group)) +  
  geom_col(position = "dodge")
```

3. Regression Analysis and Dimensionality Reduction

Principal Component Analysis (PCA)

PCA is used to reduce the dimensionality of datasets with many correlated numeric predictors.

- **Ordering:** PC1 always explains the most variance, followed by PC2, etc.
- **Loadings:** Tell you which original variables contribute most to each component.
- **Standardization:** Predictors should be standardized (scaled) before PCA if they are on different units.

Interpreting Log-Transformed Models

When the response variable is log-transformed (e.g., `log(sales)`), the interpretation of a coefficient β changes:

A one-unit increase in the predictor X is associated with a $100 \times (e^\beta - 1)\%$ change in the expected value of the original response Y .

Multiple Regression Benefits

Multiple regression allows you to estimate the effect of one predictor while **adjusting for or holding constant** the effects of other variables in the model.

4. Classification and Model Evaluation

The Confusion Matrix

For a binary classification (e.g., “Upgrade” vs. “No Upgrade”), you must know these metrics:

Metric	Formula	Interpretation
Accuracy	$(TP + TN) / \text{Total}$	Overall correctness of the model.
Precision	$TP / (TP + FP)$	Of all predicted positives, how many were correct?
Recall	$TP / (TP + FN)$	Of all actual positives, how many did we catch?
F1 Score	$\frac{2 \cdot \text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}}$	The harmonic mean of precision and recall.

Random Forest vs. Decision Trees

A **Random Forest** is generally superior to a single decision tree because it averages the results of many trees. This reduces **variance** and makes the model less sensitive to noise in the training data, thereby reducing overfitting.

5. Machine Learning Patterns in `tidymodels`

The exam expects you to write or interpret R code using the `tidymodels` framework.

Common Model Specifications

- **k-Nearest Neighbors (kNN):**

```
nearest_neighbor(neighbors = 7) %>%  
set_engine("kkn") %>%  
set_mode("regression")
```

- **Decision Trees:**

```
decision_tree() %>%  
set_engine("rpart") %>%  
set_mode("classification")
```

Key Concepts in Model Training

- **Standardization:** Mandatory for **kNN** because it relies on distance calculations. Variables with larger scales would otherwise dominate the model. It is **not necessary** for Decision Trees.
- **RMSE (Root Mean Square Error):** Measures the average error in predictions. Lower is better.
- **Overfitting Check:** Compare Training RMSE vs. Test RMSE. A large gap indicates overfitting.
- **Bias-Variance Trade-off (k in kNN):**
 - **Small k:** Low bias, high variance (Overfitting/Sensitive to noise).
 - **Large k:** High bias, low variance (Underfitting/Too smooth).

6. Exam Strategy: Prediction Risks

When asked about the reliability of a prediction for a new data point, always check for **extrapolation**. If the new data point's values (e.g., `Duration = 5`) are far outside the range of the training data (e.g., `Duration` range was 20-60), the prediction is highly unreliable because the model hasn't "learned" the relationship in that region.